

Kiera McCormick

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RESEARCH INTERESTS

Large language models for scientific and real-world understanding, with an emphasis on retrieval-augmented generation, prompt engineering, and question answering. I am especially interested in human–AI comparison and cooperation on NLP tasks, the evaluation of human–computer interaction in high-stakes and scientific settings, and the benchmarking of LLM behavior. Broader interests include open science, interdisciplinary applications of machine learning, machine translation, and AI safety.

EDUCATION

- Johns Hopkins University** Baltimore, MD
M.S. in Computer Science — Human Language Technology Expected May 2027
- **Coursework:** NLP for Computational Social Science · AI-Enabled Software Engineering · Machine Social Intelligence · Natural Language Processing · Human Language Technology
- Loyola University Maryland** Baltimore, MD
B.S. in Engineering — Electrical & Computer Engineering; Math Minor May 2025
- GPA 3.745 · Dean's List, all eight semesters (Fall 2021 – Spring 2025)
 - **Coursework:** Signals & Systems · Digital Logic · FPGA Design · Microprocessor-Based Systems · Quantum Computing I & II · Linear & Advanced Linear Algebra · Probability & Statistics · Communications
- Danish Institute for Study Abroad (DIS)** Copenhagen, DK
Core Course: Holocaust and Genocide Spring 2024

RESEARCH & WORK EXPERIENCE

- JHU Human Language Technology Center of Excellence (HLTCOE)** Baltimore, MD
Research Intern, Summer Camp for Applied Language Exploration (SCALE) June 2026 – Present
- Contributing to a summer research program on evaluation methods for multimodal retrieval-augmented generation (RAG) systems.
 - Developed a novel evaluation metric that better captures how well AI systems tailor information to different user needs, addressing limitations of existing approaches.
 - Validated the metric through human-agreement studies and empirical testing across multiple real-world event datasets.
- Johns Hopkins University** Baltimore, MD
Research Assistant — AI for Forensics (with NIST) January 2026 – Present
- Conducting research at the intersection of AI and forensic science with the National Institute of Standards and Technology (NIST).
 - Surveying AI tools across forensics-adjacent domains and building structured analyses of AI adoption patterns.
 - Contributing to stakeholder documentation on responsible AI integration in forensic laboratories.
- Center for Astrophysics | Harvard & Smithsonian** Cambridge, MA
AstroAI Summer Intern January 2025 – August 2025
- Engineered prompts to generate concise, discriminative summaries of astrophysical X-ray sources from Chandra archival data, with visualizations to characterize prompting effects.
 - Extracted embedded physical parameters to evaluate LLMs' grasp of scientific knowledge.
 - Applied sparse autoencoders to surface novel patterns and relationships among astrophysical objects.
- Space Telescope Science Institute & Johns Hopkins University** Baltimore, MD
Summer Astronomy Program / Jelinek Memorial Summer Workshop May 2024 – August 2024
- Investigated applications of machine learning and LLMs in astronomy research, assessing their potential to improve research efficiency and reliability.
 - Designed and deployed LLM-powered Slack chatbots, including prompt engineering, open- vs. closed-source model comparisons, information retrieval pipelines, and evaluation benchmark datasets.
- Omega Technical Services at Los Alamos National Laboratory** Los Alamos, NM
Engineering Intern May 2023 – August 2023
- Automated a previously manual job-scheduling process for the industrial engineering team, replacing it with a database-driven system.
 - Built metrics and dashboards using MicroStrategy, SQL, Python, VBA, and Excel; developed an ETL pipeline with MariaDB, MS Task Scheduler, and Alteryx.

Loyola University Maryland

Physics Teaching Assistant & Tutor

Baltimore, MD

September 2022 – May 2025

- Supported Physics I and II instruction through problem-solving sessions and individual tutoring, emphasizing quantitative reasoning and conceptual understanding.

University of Delaware

Junior Physics Intern

Newark, DE

March 2020 – August 2020

- Analyzed the spectral characteristics and luminosity of massive stars; developed Python/Jupyter scripts to process and visualize stellar data.
- Created visualizations of spectral lines and magnitude measurements to aid interpretation of stellar properties.

PUBLICATIONS

McCormick, K., & Martínez-Galarza, J. R. (2025). *Encoding and Understanding Astrophysical Information in Large Language Model-Generated Summaries*. NeurIPS Workshop on Machine Learning and the Physical Sciences (ML4PS) — accepted.

McCormick, K., Hyk, A., Zhong, M., Ciucă, I., et al. (2025). *From Queries to Criteria: Understanding How Astronomers Evaluate LLMs*. Conference on Language Modeling (COLM) — accepted.

Wu, J. F., Hyk, A., **McCormick, K.,** Ye, C., et al. (2024). *Designing an Evaluation Framework for Large Language Models in Astronomy Research*. arXiv:2405.20389.

Iyer, K. G., Yunus, M., O'Neill, C., Ye, C., Hyk, A., **McCormick, K.,** et al. (2024). *pathfinder: A Semantic Framework for Literature Review and Knowledge Discovery in Astronomy*. ApJS, 275, 38; arXiv:2408.01556.

TALKS & POSTER PRESENTATIONS

“Encoding and Understanding Astrophysical Information in LLM-Generated Summaries” — STScI: Language AI in the Space Sciences, Baltimore, MD. Poster, March 2026.

“Encoding and Understanding Astrophysical Information in LLM-Generated Summaries” — NeurIPS ML4PS Workshop, San Diego, CA. Poster, December 2025.

“From Queries to Criteria: How Astronomers Evaluate LLMs” — Conference on Language Modeling (COLM), Québec, Canada. Poster, October 2025.

“Evaluating Large Language Models in Astronomy Research” — AstroAI Workshop, Cambridge, MA. Poster, July 2025.

“Evaluating Large Language Models in Astronomy Research” — American Astronomical Society 245, National Harbor, MD. iPoster, January 2025.

“Evaluating LLM Tools: Insights from Astronomer Interactions with a RAG-Based Slack Chatbot” — AAS 245, National Harbor, MD. Co-authored talk, January 2025.

“pathfinder: A Semantic Framework for Literature Review and Knowledge Discovery in Astronomy” — AAS 245, National Harbor, MD. Co-authored iPoster, January 2025.

“Applications of Large Language Models in Physics and Astronomy” — Gender Minorities and Women in Physics Summit, JHU. Poster, September 2024.

PROFESSIONAL SERVICE & OUTREACH

- Reviewer**, NeurIPS EvalEval Workshop (2024).
- Booth Staff**, NIST booth, American Society of Crime Laboratory Directors, Grand Rapids, MI (May 2026) — gathered practitioner perspectives on AI adoption in forensic science.
- Volunteer**, AstroAI Workshop.
- Completed the 2024 JHU Summer School on Human Language Technology.

SELECTED PROJECTS

A2CPS Discovery Agent

Jan – May 2026

- Co-developed a natural-language interface for querying the A2CPS biomedical dataset, translating plain-English questions into read-only SQL.
- Built a retrieval-augmented generation pipeline surfacing relevant study documentation alongside query results; implemented features in a React/Hono application using the Anthropic Claude API.

Regional Variation in ChatGPT Use Across States

Jan – May 2026

- Applied BERTopic to 68,668 first-turn prompts from the WildChat corpus to study geographic and socioeconomic variation in ChatGPT use across 36 states.
- Found that higher education and income correlate with specialized creative uses, while generic image-generation requests are more common in lower-education, lower-income states.

Testing Social Reasoning in LLMs

Jan – May 2026

- Designed a multi-dimensional benchmark of 40 original scenarios spanning Theory of Mind, Pragmatics, and Chain-of-Thought reasoning across 12 state-of-the-art LLMs.
- Recruited a human baseline (n = 12) to quantify the gap between model and human-level social reasoning.

Hotspot Homing Robot

Sep 2024 – May 2025

- Developed an autonomous mobile robot to detect and localize rogue wireless access points in office environments (sponsored and mentored by Raytheon Applied Signal Technology).
- Integrated wireless signal processing with SLAM-based navigation and a real-time threat-visualization interface.

TECHNICAL & RESEARCH SKILLS

Programming: Python · SQL/SQLite · Java · C++ · Git/GitLab · HTML

ML / AI: PyTorch · TensorFlow · LLMs · retrieval-augmented generation & information retrieval · representation learning · model evaluation & benchmarking · prompt engineering

Scientific / Engineering: MATLAB (certified) · Simulink · LTspice · Active-HDL · MicroStrategy

Hardware / Systems: Raspberry Pi · Arduino · FPGA · embedded systems

LEADERSHIP & ACTIVITIES

Academic All-American — JHU Women's Club Lacrosse

Sep 2025 – Present

President — Society of Women Engineers, Loyola

Feb 2023 – May 2025

Class of 2025 Representative — Engineering Industrial Advisory Board

Dec 2023 – May 2025

Member & Peer Mentor — Loyola Honors Program · Haig Scholar

Sep 2021 – May 2025

Academic All-American — Loyola Women's Club Lacrosse

Sep 2022 – May 2025

Member — Institute of Electrical and Electronics Engineers (IEEE)

Oct 2021 – May 2025